

Module 6

Outcome Attainment

Hands-on Training on Outcome-Based Education
(OBE)

Board of Accreditation for Engineering and Technical Education (BAETE)
Training on OBE and Accreditation, 26-27 June 2026



Presented by

Mafin Muntasir Rahman
Assistant Professor
Dept. of EEE, BAUST

Courtesy of

Dr. Sadid Muneer
Director, IR and Industry Liaison &
Member Secretary, Training Taskforce, BAETE
Associate Professor, EEE, UIU



Outline

- Directives on Outcome Attainment in BAETE Manual and SAR Template
- CO Attainment
- PO Attainment
- PEO Attainment
- Continuous Quality Improvement (CQI)
- Summary

Main Components of OBE

Course Outcomes (CO)

COs address the abilities to be attained by students upon completion of a course.

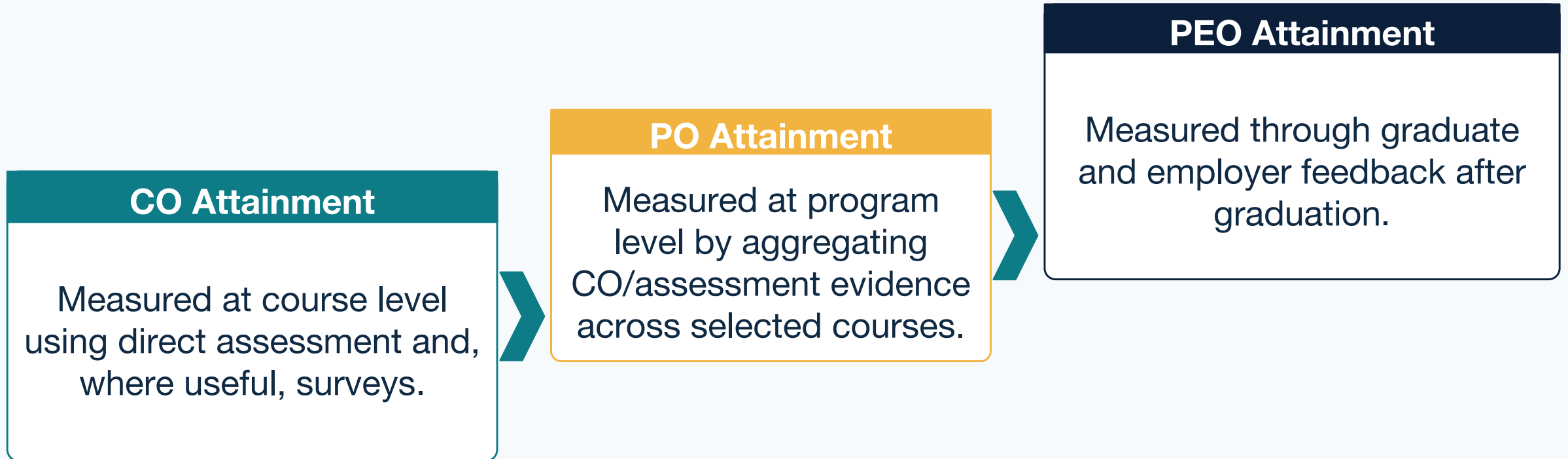
Program Outcomes (PO)

POs describe what students are expected to know and be able to perform by the time of graduation.

Program Educational Objectives (PEO)

PEOs address graduates' attainment around 3-5 years after graduation.

Attainment of Outcomes



Outline

1. Directives on Outcome Attainment in BAETE Manual and SAR Template

2. CO Attainment

3. PO Attainment

4. PEO Attainment

5. Continuous Quality Improvement (CQI)

6. Summary

From Manual / SAR Template

- **SAR Template, ACC-TMP-04-04-V3-F, 2.4 Assessment of Course Outcomes:** Describe how the attainment of COs is assessed in each course, including rubrics where applicable. **Assessment processes, attainment criteria, scale, and expected attainment level should be clearly stated.**
- **Accreditation Criteria, ACC-MAN-02-v3-F, 5.2 Program Outcomes and Assessment:** The program must demonstrate, **using direct methods**, that students attain all POs by graduation.

From Manual / SAR Template

- **Accreditation Criteria, ACC-MAN-02-v3-F, 5.1 Program Educational Objectives:** The program must demonstrate an established system to compile PEO attainment including a mechanism for tracking and to obtain feedback from graduates and their employers.
- **SAR Template, ACC-TMP-04-04-V3-F, 5.4.2 CQI Loop for PO:** Describe the CQI processes for POs. **Discuss how direct and indirect assessment results, including feedback from stakeholders, are systematically used. Evidence should show that PO attainment is continuously improved.**

Assessment Methods

- Direct assessment methods in courses.
- Indirect assessment methods through surveys and feedback.

Assessment through Surveys

Assessment Area	Typical Survey/Feedback	Purpose
CO assessment	• End-of-course student surveys • Need to update course evaluation questionnaire	Find gaps in course delivery and assessment
PO assessment	• Exit survey for all graduating students • Questions to address all 12 POs	Support program-level attainment evidence
PEO assessment	• Graduate and employer surveys • Questions must address all PEO statements • Graduates 3-5 years after graduation	Assess long-term graduate outcomes

Outline

1. Directives on Outcome Attainment in BAETE Manual and SAR Template

2. CO Attainment

3. PO Attainment

4. PEO Attainment

5. Continuous Quality Improvement (CQI)

6. Summary

CO Assessment in Courses

- Assessment of CO attainment and assessment of grades in a course are different.
- Which courses should we consider?
 - CO assessment in all courses?
 - May be for selective section in case of multiple sections
 - May be in selective semester/trimester?
 - Assessment in selective core courses?
- Even in case of sampling, CO attainment in all courses is required for course-level CQI.
- **SAR Template, ACC-TMP-04-04-V3-F, 5.4.3 CQI Loop for Courses and Curriculum:** Describe the CQI processes for courses and curriculum. Discuss how different stakeholders' assessment results and feedback are used to continuously improve COs, attainments, and curriculum.

CO Assessment in Theory Courses

- May exclude class tests/quizzes
 - most of the time not all CTs are counted (3 out of 4)
 - Midterm and final examinations usually cover the whole syllabus
 - No need to include the same question/content twice
 - Too many assessments will make the process difficult
 - Assignments are useful for open-ended, individual problems
- First, we need to prepare a course syllabus that must have
 - COs
 - CO assessment methods
 - CO-PO mapping

Example CO-PO Mapping of a Theory Course (Electronics)

CO	Description	PO	K/P/A	Bloom Domain and level	Delivery Methods and Activities	Assessment Tools
CO1	Explain behavior and functionality of basic electronic devices and circuits with those devices.	PO1	K3	Cognitive / Understand	- Lecture - Problem solving - Group activities	- Quiz - Class Test - Exam - Assignment
CO2	Determine parameters for electronic circuits indicative of performance.	PO2	K3, P1	Cognitive / Apply		
CO3	Apply concepts of electronic devices and Op-Amps to design/construct circuits.	PO3	K5, P1, P2, P3	Cognitive / Apply		
CO4	Interpret frequency response of a circuit with Bode plots.	PO1	K3	Cognitive / Apply		

Example of Assessments in Theory Course

No.	Assessment Item	Marks Allocated	CO1	CO2	CO3	CO4
			PO1	PO2	PO3	PO1
1	Attendance	5%				
2	Class test	15%				
3	Assignment	15%			✓	
4	Mid-term exam	30%	✓	✓	✓	✓
5	Final exam	35%	✓	✓	✓	✓

Mapping of Questions to COs

- All assessments (CT, assignments, mid, final) that are considered must map to the COs.
- For mid/final, mapping can be included with question moderation.
- Fill up your own question moderation form.
- This will clearly indicate whether you are maintaining balance in the question.

COs	Mid	Final	Assignment
CO1	Q1	Q3	
CO2	Q2	Q1	
CO3	Q4/Q5	Q2	✓
CO4	Q3	Q5/Q6	

Mapping of Questions to COs

- Keep the work as simple as possible.
- **Note, All questions need not be mapped.**
- Questions that are mapped should be compulsory.
- Options may be given if all questions map to the same CO.
- One question should not map to multiple COs.
- Avoid too many breakdowns/details of mapping.

COs	Mid	Final	Assignment
CO1	Q1	Q3	
CO2	Q2	Q1	
CO3	Q4/Q5	Q2	✓
CO4	Q3	Q5/Q6	

CO Assessment in Lab Courses

- Not as simple as theory courses.
- Assessment components may vary
 - Mid/Final exam (quiz)
 - Software/Hardware test
 - Report
 - Viva
 - Presentation
 - Individual/group project
 - Weekly programming or hardware assignments.
- No need to bring everything into CO assessment.

CO Assessment in Lab Courses: Suitable POs

POs	Assessment
PO4: Investigation	Software / hardware open-ended projects
PO5: Modern Tool Usage	Tools taught in lab classes
PO8: Individual and Team Work	Group projects
PO9: Communication	Viva, project presentation through slides, posters, written reports etc.

Example of CO Assessment in a Lab Course

CO	Description	PO	K/P/A	Bloom's level	Delivery Methods / Activities	Assessment Tools
CO1	Analyze behavior/response of electronic circuits using modern tools.	PO 2	K3, P1	Cognitive / Apply	• Circuit simulation • Circuit implementation using discrete components • Real-life problem solving	• Pre-lab assessment • Class performance • Mid • Demonstration
CO2	Design and implement electronic circuits meeting specific requirements.	PO 3	K3, K5, P1, P2, P3	Cognitive / Apply	Term project	Project demonstration
CO3	Prepare and present a technical report.	PO 9	A1, A2	Psychomotor	• Data collection using computer software • Report writing	• Lab report • Project report • Presentation
CO4	Work individually and in a group to complete a project.	PO 8		Affective	• Term project • Group activities	Project demonstration

Example of CO Assessment in a Lab Course

No.	Assessment Item	Marks Allocated	CO1	CO2	CO3	CO4
			PO2	PO5	PO10	PO3
1	Active class participation	30%				
2	Laboratory report	10%			✓	
3	Mid-term evaluation	20%	✓			
4	Project	40%	✓	✓	✓	✓

Assessment Rubrics

Prior Practice (should be avoided)	Rubric-Based Practice
No clear marking scheme for project, presentation, viva or report.	Provides a clear breakdown for project, presentation, viva, report and similar tasks.
Marks may be based only on overall impression, e.g., 8/9 for good work and 4/5 for unfinished work.	Provides transparent criteria and performance levels.
Students may not know exactly what is expected.	Informs students 'What We Want' from them in presentations, projects and reports.
Often difficult to justify marks consistently.	Mainly useful for labs or theory courses with project/presentation components.

Use of Rubrics

Rubrics are required for the following assessments:

- Report
- Viva
- Presentation
- Individual / group project
- Project report
- Weekly programming / hardware assignments

Assessment Rubrics of a Lab Project

Assessment Item	CO/PO	Criteria / Remarks	K/P/A
Abstract submission (10)	CO1-PO2	Submit on time; penalties will incur if there is delay in submission.	K3, K8, P1, P2
Breadboard demonstration (20)	CO2-PO3	Project works according to proposal in breadboard.	K5, P2, P3
Veroboard demonstration (20)	CO2-PO3	Project works according to proposal in veroboard.	K5, P2, P3
Individual viva (10)	CO4-PO8	Questions on theory and implementation asked individually.	K5, K8
Presentation (10)	CO3-PO9	Individual presentation skill judged.	A1, A2
Report (10)	CO3-PO9	Title page; Introduction & Conclusion (2); Description (2); Circuit diagram & Explanation (2); Simulation Results (2); Picture of Project (1); Results from Project (1).	A1 (Project Description), A2 (Circuit Diagram & Explanation)

Assessment of CO Attainments

- **SAR Template, ACC-TMP-04-04-V3-F, 2.4 Assessment of Course Outcomes:** Describe how CO attainment is assessed in each course, including rubrics where applicable. The assessment processes, attainment criteria, scale, and expected level of attainment should be clearly stated.
- Attainment criteria and scale must be defined.

Assessment of CO Attainments

How to determine that a CO is attained for an individual student and for a course?

Individual Student

- Needed for PO attainment calculation.
- Program sets the criterion.
- **Example:** if a student receives 55% **(set by Program)** of marks assigned for a CO in a course, the student attains that CO.

Course Level

- Needed for course-level CQI.
- Program sets the criterion.
- **Example:** if 50% **(set by Program)** of students in a course attain a CO, that CO is attained for the course.

CO Attainment Example

Assigned marks: 100 for each CO

Roll	Name	CO1	CO2	CO3	CO4
1	Student A	7	21	11	0
2	Student B	94	56	51	77
3	Student C	38	37	8	20
4	Student D	57	46	24	21
5	Student E	75	26	84	92
6	Student F	25	25	17	88
7	Student G	88	82	100	93
8	Student H	69	35	62	87
9	Student I	69	73	83	87
10	Student J	88	89	100	74
% attained		70%	40%	50%	70%

Criterion 1: Individual Student

If a student attains 55% of the total marks assigned to a particular CO, that CO is attained.

Criterion 2: Course Level

If 50% of students attaining a CO means that CO for the course is attained.

Outline

1. Directives on Outcome Attainment in BAETE Manual and SAR Template
2. CO Attainment
- 3. PO Attainment**
4. PEO Attainment
5. Continuous Quality Improvement (CQI)
6. Summary

POs: CEP and CEA

No.	Program Outcome	CEP / CEA
1	Engineering Knowledge	CEP
2	Problem Analysis	CEP
3	Design / Development of Solutions	CEP
4	Investigation	CEP
5	Modern Tool Usage	CEP
6	Environment and Sustainability	CEP
7	Ethics	-
8	Individual and Team Work	-
9	Communication	CEA
10	Project Management and Finance	-
11	Life-Long Learning	-
12	Business Idea	-

Assessment of PO Attainment

- **CEP is already embedded in the definition of some POs.**
- **Example** - PO1: Apply knowledge of mathematics, natural science, engineering fundamentals and engineering specialization to solve complex engineering problems.
- **SAR Template, ACC-TMP-04-04-V3-F, 2.7.1 Attainment of PO1 or its Equivalent:** Describe how PO1 or its equivalent is assessed and evaluated.
 - State and justify methods, tools, criteria and scale used in assessment process.
 - State how WKs are incorporated (K1-K4)
 - identify which CEP (P1-P4) are addressed through the attainment of PO1 and provide supporting evidence.
 - Present a summary of results obtained after the assessment and analysis to demonstrate the extent to which PO1 is being attained.

Demonstrating Attainment of CEP and CEA

- For POs with CEP/CEA in their definition
 - Demonstrate PO attainment through CEP/CEA evidence
- Different assessment tools can be used:
 - Project in lab/theory courses
 - Assignment
 - Presentation
 - Report
 - Final exam question
- Assessment tools used to assess these POs must be demonstrated as CEP/CEA through proper mappings.

Things to Consider before the Assessment of PO Attainments

- **SAR Template, ACC-TMP-04-04-V3-F, 2.7 Attainment of Program Outcomes:**

Provide a summary of the results of the attainment of program outcomes for the most recent graduating cohort, whose attainment was measured.

Things to Consider before the Assessment of PO Attainments

- **Fix the definition of cohort**

- Trivial for fixed-credit programs
- Graduating/entry semester can be used for open-credit programs

- **Use a robust, scalable system**

- Use Excel files or software (if available)
- Prepare Excel file with detailed formulas

- **Course Selection to show attainment of all 12 POs**

- Final Year Design Project/Capstone may be selected, but relying on it alone is not easy; include more courses
- Third-year and final-year courses are usually more suitable

Things to Consider before the Assessment of PO Attainments

- **CO-PO mapping**

- One-to-one mapping is always easier to implement
- Add more COs if needed to convert mapping into one-to-one

- **Different weightages in CO-PO mapping**

- Introduce complicated calculations and are not always necessary

- **Importance of Rubrics**

- Rubrics are very important, especially if a single assessment tool is used for multiple POs (e.g., a report used to assess multiple POs)

Example of PO Attainments for PO(a)/PO1

- PO attainment threshold must be defined.
- **Example:** if a student gets 50% or more, then the PO is considered attained.
- A simple equal-level model is easier to manage.
- Using a complex weighted model requires additional complexity in curriculum design and PO assessment.

Course	CO	Domain
CSE 3325	CO1	C
CSE 3325	CO2	C
CSE 3214	CO3	C
CSE 4135	CO2	C

Equal Level
% of marks obtained in different COs is calculated considering equal level.

Course	CO	Domain	Level
CSE 3325	CO1	C	2
CSE 3325	CO2	C	3
CSE 3214	CO3	C	1
CSE 4135	CO2	C	2

Weighted Level
% of marks obtained in different COs is calculated considering different levels. A higher level contributes more to calculating %.

PO Calculation (Example)

Course	CO-PO Mapping			
Course 1	CO1-PO1	CO2-PO3	CO3-PO5	CO4-PO9
Course 2	CO1-PO1	CO2-PO2	CO3-PO4	CO4-PO8
Course 3	CO1-PO1	CO2-PO3	CO3-PO6	CO4-PO10

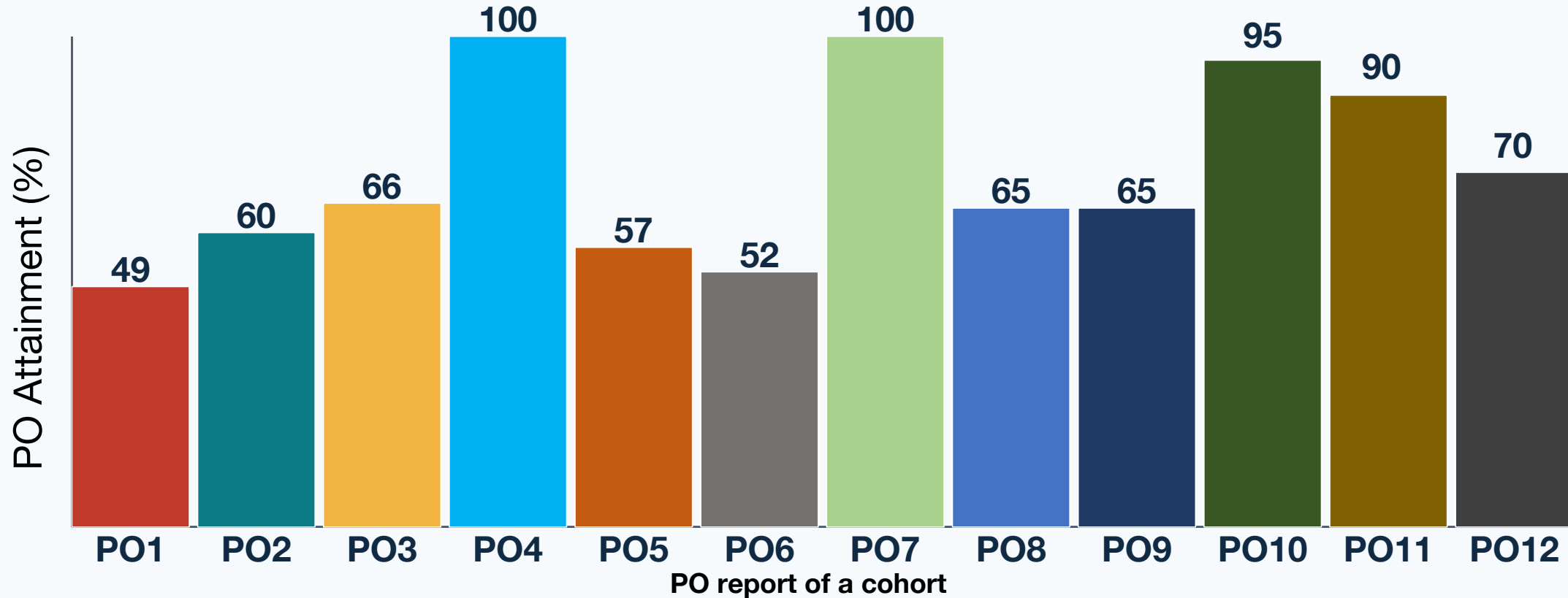
Calculation

Calculate PO1 attainment from courses 1, 2 and 3 for a cohort

Criterion for PO attainment

If the PO assessment score is greater than 50%, the PO is attained. **The actual criterion should be set by the program.**

PO Calculation (Example)



Observations

1. Only PO1 was not attained for this cohort.
2. Other POs were attained.

Outline

1. Directives on Outcome Attainment in BAETE Manual and SAR Template
2. CO Attainment
3. PO Attainment
- 4. PEO Attainment**
5. Continuous Quality Improvement (CQI)
6. Summary

Assessment of PEO Attainment

- **Accreditation Criteria, ACC-MAN-02-V3-F, 5.1 Program Educational Objectives:** the program must demonstrate an established system for compiling the level of attainment of PEOs. The system should include tracking and feedback from graduates and employers.
- **Important stakeholders: employers and alumni.**
- **Survey is the preferred tool.**
 - Questions should be designed for each PEO statement.
 - How survey responses are used to assess PEO attainment must be indicated.
- **If there is no OBE-based graduating batch, PEO can be assessed with a non-OBE batch following BAETE guidance.**

Outline

1. Directives on Outcome Attainment in BAETE Manual and SAR Template
2. CO Attainment
3. PO Attainment
4. PEO Attainment
- 5. Continuous Quality Improvement (CQI)**
6. Summary

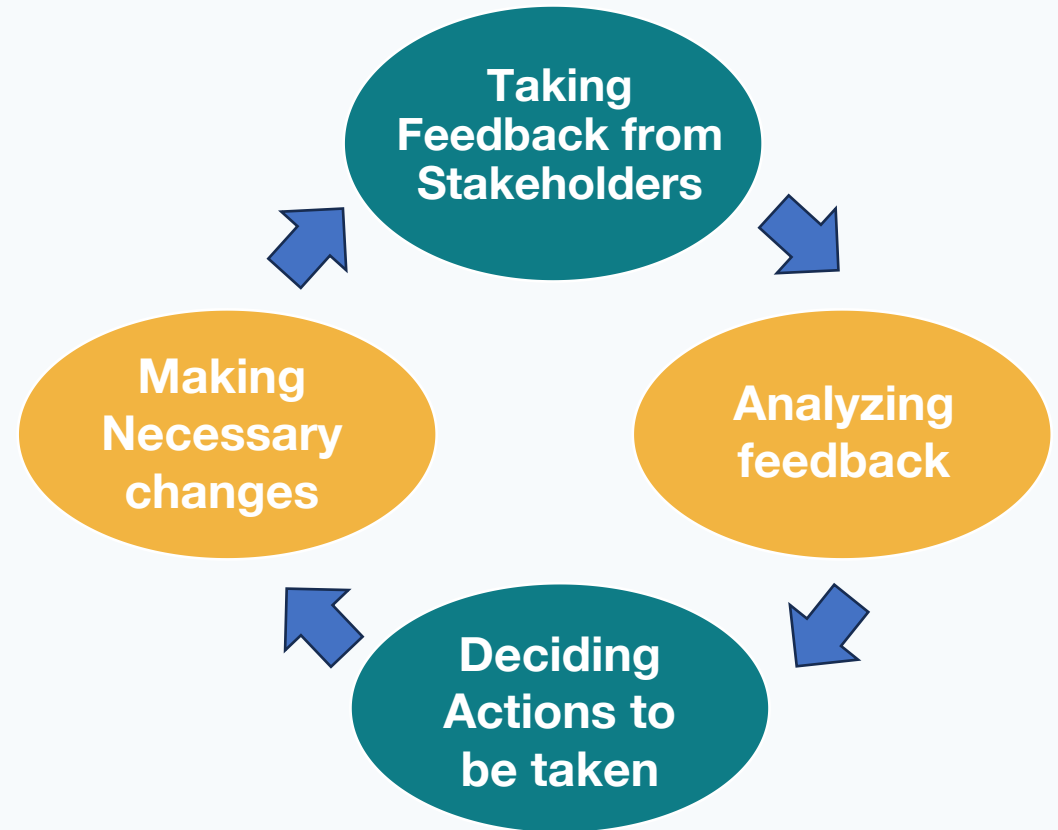
Close the Loop through Continuous Quality Improvement (CQI)

- **Establish a CQI cycle**

- define the period after which actions will be taken for improvement.
- For CO, PO and PEO, different durations may be used.

- **Identify stakeholders**

- to collect feedback
- Students, faculty members, employers and alumni are popular stakeholders
- Select relevant stakeholder groups for CO, PO and PEO



Continuous Quality Improvement (CQI)

The program must complete the CQI loops periodically

CQI Area	Primary Feedback Source
COs	Course instructors and students (secondary)
POs	Relevant stakeholders, including graduates
PEOs	Graduates and their employers
Curriculum and teaching quality	Faculty members and students

CQI loops should be a regular practice - not something done only before an accreditation visit.

Summary

- The program needs to assess attainment of COs in each course.
- Rubrics, assessment processes, attainment criteria, scale, and expected level of attainment should be clearly defined.
- The program must demonstrate, using direct methods, that students attain all POs by graduation.
- A system for compiling PEO attainment must be established. Feedback from graduates and their employers need to be collected.
- Assessment of outcome attainment is vital for CQI.

End of Module

Module 6: Outcome Attainment



Thank You